

Phase 1:

Unnormalized Table

Before normalization, all FitZone gym data could have been stored in one large table that combines members, sessions, classes, trainers, and feedback information together.

Example of an unnormalized table:

ID	Member_Name	Email	Phone	Session_ID	Session_Date	Start_Time	Class_Na
1	John Smith	john@email.com	123456789	1	2025-04-27	08:00	Beginne Yoga
2	John Smith	john@email.com	123456789	2	2025-05-01	12:00	Beginne Yoga
3	Emma Johnson	emma@email.com	987654321	3	2025-05-03	09:00	Power Yo

Problems with the Unnormalized Table

The unnormalized table contains several problems because multiple types of information are stored together.

1. Data Duplication

Member and trainer information would be repeated every time a member books a session.

For example, if John Smith books five sessions, his name, email, and phone number would appear five times. Similarly, trainer details would be repeated for every session they teach. This wastes storage space and increases the chance of inconsistent data.

Example:

ID	Member_Name	Email	Session_ID
1	John Smith	john@email.com	1
2	John Smith	john@email.com	2
3	John Smith	john@email.com	5

2. Update Anomalies

Because data is duplicated, updating information becomes difficult.

For example, if a trainer changes their phone number, every row containing that trainer's information must be updated. If one row is missed, the database will contain conflicting information.

Example:

Before update:

Trainer_ID	Trainer_Name	Phone
2	Jessica Lee	111111111
2	Jessica Lee	222222222

The same trainer has two different phone numbers because not all records were updated.

3. Insertion Anomalies

The unnormalized table makes it difficult to insert new data.

For example, adding a new trainer who has not taught any sessions yet would be impossible because the table requires session information as well.

A trainer should be able to exist independently from sessions.

4. Deletion Anomalies

Deleting data about one activity may accidentally remove other important information.

For example, if the only booking for a specific trainer is deleted, the trainer's information could also disappear from the table.

5. Repeating and Multi-valued Attributes

The rating and comment columns may contain multiple values because a member can book multiple sessions and leave feedback for each booking.

Example:

ID	Session_ID	Ratings	Comments
1	1	5,4,3	Excellent, Good, Average

This violates the first normal form (1NF), which requires each field to contain a single atomic value.